

Quiz 2

Subject: MLT 2025 Mar16: IIT M AN EXAM QDF4 Total Marks: 40.00 Questions: 17

QUESTIONS

Q1

MCQ

[0.00 marks]

THIS IS QUESTION PAPER FOR THE SUBJECT "**DIPLOMA LEVEL : MACHINE LEARNING TECHNIQUES (COMPUTER BASED EXAM)**" **ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?** CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN. **(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)**

OPTIONS

A. YES ✓ **Correct**

B. NO

Below is the constrained version of the ridge regression optimization problem:

$$\min_{w \in \mathbb{R}} \sum_{i=1}^n (w^T x_i - y_i)^2$$

$$\text{subject to } \|w\|^2 \leq \theta.$$

Following are the weight vectors to be considered, along with the mean squared error (MSE) produced by each:

$$w_1 = [1 \ 1 \ 1 \ 1]^T, \text{ MSE} = 2$$

$$w_2 = [0 \ 2 \ 1 \ 3]^T, \text{ MSE} = 7$$

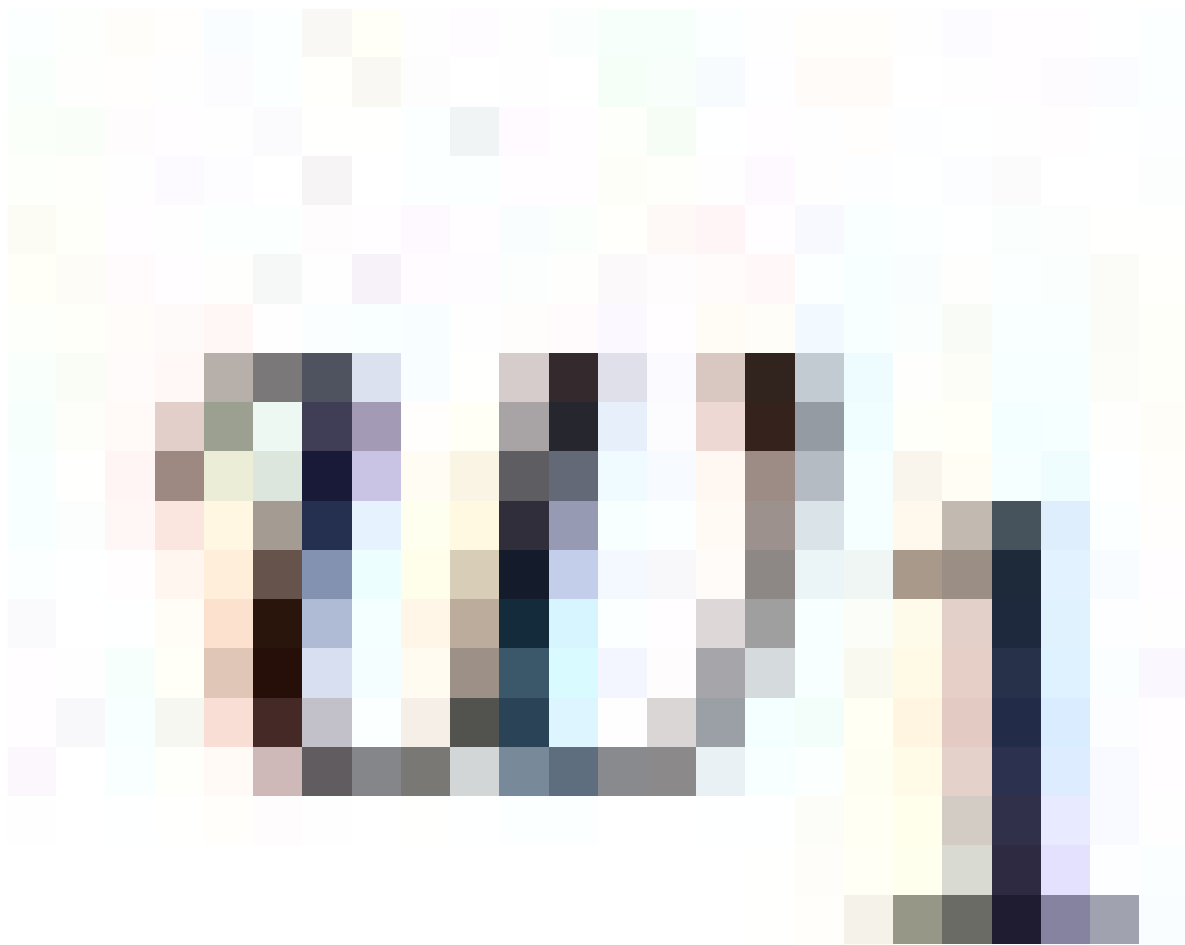
$$w_3 = [1 \ 2 \ 0 \ 1]^T, \text{ MSE} = 1$$

$$w_4 = [2 \ 1 \ 1 \ 2]^T, \text{ MSE} = 8$$

If $\theta = 10$, which of the following weight vectors will be selected as the final weight vector by ridge regression?

OPTIONS

A.



B.



C.



✓ Correct

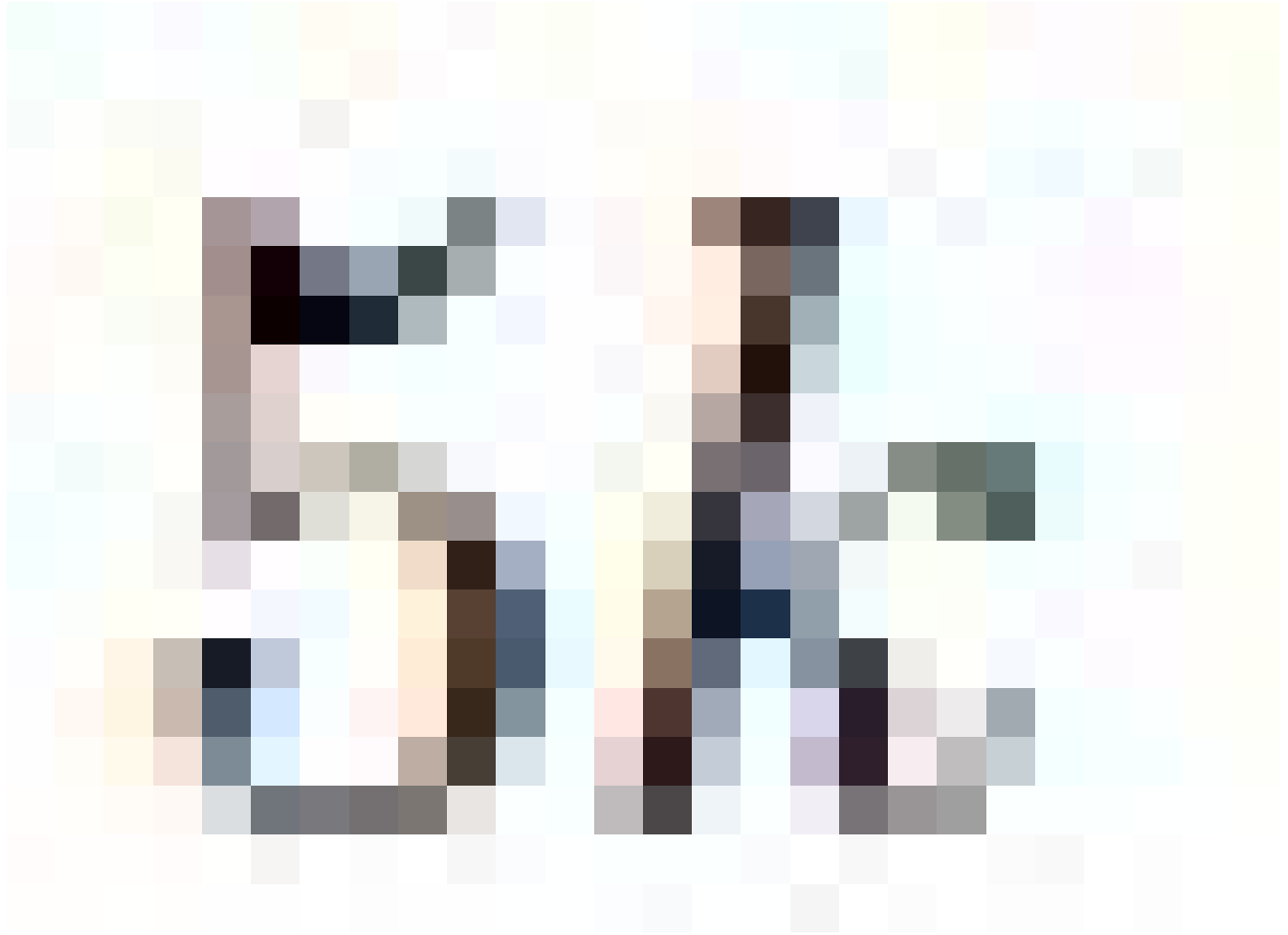
D.



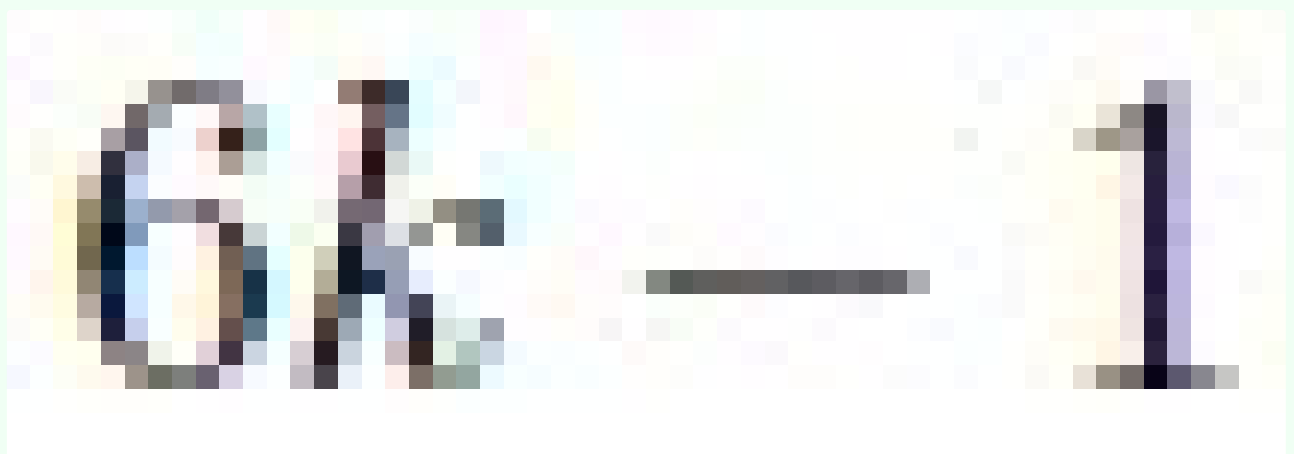
Consider a k - class classification problem with five binary features f_1, f_2, f_3, f_4 , and f_5 , where $k > 2$. How many parameters need to be estimated to learn a Naive Bayes classifier?

OPTIONS

A.



B.



✓ Correct

C.



D.



Q4 MCQ

[2.00 marks]

p is the proportion of points with label 1 in some node in a decision tree. Which of the following statements are true?

OPTIONS

- A. As the value of p increases from 0 to 1, the impurity of the node increases.
- B. As the value of p increases from 0 to 1, the impurity of the node decreases.
- C. The impurity of the node does not depend on p .
- D. $p = 0.5$ correspond to the case of maximum impurity. ✓ **Correct**

What is the main purpose of using Laplace smoothing in Naïve Bayes?

OPTIONS

- A. To prevent numerical errors when calculating the variance of Gaussian features.
- B. To ensure that unseen categorical feature values receive small but non-zero probabilities. ✓ **Correct**
- C. To strengthen the model's predictions by increasing the weight of certain features.
- D. To balance the likelihood values of all features before computing posterior probabilities.

Consider the following feature (x) along with its label (y):

x	y
1	1
-1	1
2	1
-2	0
3	0
4	1

If you are provided with the following two questions, which one should you choose for the first (top) node of the decision tree?

- (a) $x \leq -2$
- (b) $x \leq -1$

Recall: Information gain = $\text{Entropy}(D) - \gamma \text{Entropy}(D_{Yes}) - (1 - \gamma) \text{Entropy}(D_{No})$

Use the following values if required:

$\log_2(3/4) = -0.41$, $\log_2(4/5) = -0.32$, $\log_2(1/5) = -2.32$, $\log_2(1/4) = -2$, $\log_2(1/2) = -1$.

OPTIONS

- A. (a) ✓ **Correct**
- B. (b)
- C. Insufficient information

Polynomial kernel in kernel regression model is expressed as:

$$K(x, x') = (x^T x' + 1)^d$$

What role does $K(x, x')$ play in kernel regression?

OPTIONS

- A. It ensures that the regression model remains linear.
- B. It computes the similarity between data points in a transformed feature space. ✓ **Correct**
- C. It allows the model to capture non-linear relationships by mapping data to a higher-dimensional space. ✓ **Correct**
- D. It helps in reducing the variance of the model.
- E. It minimizes the loss function in regression problems.

Let $\hat{w}_R$ and $\hat{w}$ be the ridge coefficient estimate and the least squares estimate, respectively, of the linear regression problem. Which of the following holds true?

OPTIONS

A.

$$\text{For } \lambda = 0, \frac{||\hat{w}_R||^2}{||\hat{w}||^2} = 1$$

✓ Correct

B.

$$\text{For } \lambda = 0, \frac{||\hat{w}_R||^2}{||\hat{w}||^2} = 0$$

C.

$$\text{For } \lambda = \infty, \frac{||\hat{w}_R||^2}{||\hat{w}||^2} = 0$$

✓ Correct

D.

$$\text{For } \lambda = \infty, \frac{||\hat{w}_R||^2}{||\hat{w}||^2} = 1$$

Consider a kernel regression problem on the following dataset with two features:

$$\{(x_1, y_1), (x_2, y_2), (x_3, y_3)\}$$

where:

$$x_1 = \begin{bmatrix} 1 & 2 \end{bmatrix}^T, y_1 = 3, \quad x_2 = \begin{bmatrix} 2 & 3 \end{bmatrix}^T, y_2 = 5, \quad x_3 = \begin{bmatrix} 3 & 4 \end{bmatrix}^T, y_3 = 7$$

The kernel function is given as:

$$K(x, x') = (1 + x^\top x')^2$$

and the optimal coefficients α_i are given as:

$$\alpha = [0.5 \quad 1.0 \quad 1.5]^T.$$

Then, what will be the predicted value for the data point $\begin{bmatrix} 2 & 2 \end{bmatrix}^T$?

ANSWER

483 (Equal • Numeric)

Consider a Naive Bayes problem with three features, f_1 , f_2 and f_3 for a binary classification problem. We have a total of 6 training samples given in the table below:

Sample	f_1	f_2	f_3	y
x_1	0	1	1	0
x_2	1	0	1	0
x_3	0	0	1	0
x_4	0	0	0	1
x_5	1	0	1	1
x_6	0	1	0	1

For $x_{test} = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^T$, what output would Naive Bayes model predict assuming that no Laplacian smoothing has been done as part of the algorithm?

ANSWER

1 (Equal • Numeric)

CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider a linear regression model $y_i = w^T x_i$ with an initial weight $w = [3]$.

x_1	y
1	4
2	7
3	10
4	13

Consider a linear regression model $y_i = w^T x_i$ with an initial weight $w = [3]$.

x_1	y
1	4
2	7
3	10
4	13

Using the squared loss function and a learning rate of $\eta = 0.2$, compute the updated weight after one iteration of gradient descent for the following dataset:

$$\text{Use } \nabla f(w) = 2(XX^T)w - 2Xy$$

ANSWER

7 (Equal • Numeric)

CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider a linear regression model $y_i = w^T x_i$ with an initial weight $w = [3]$.

x_1	y
1	4
2	7
3	10
4	13

Consider a linear regression model $y_i = w^T x_i$ with an initial weight $w = [3]$.

x_1	y
1	4
2	7
3	10
4	13

If we stop the algorithm at the weight calculated in the previous question, what will be the predicted value for the data point, $x_1 = 5$?

ANSWER

35 (Equal • Numeric)

CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider the following 2D data points along with its labels:

x	y
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 1 & -1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 2 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & -2 \end{bmatrix}^T$	0
$\begin{bmatrix} 3 & -3 \end{bmatrix}^T$	0
$\begin{bmatrix} 4 & -1 \end{bmatrix}^T$	1

Consider the following 2D data points along with its labels:

x	y
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 1 & -1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 2 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & -2 \end{bmatrix}^T$	0
$\begin{bmatrix} 3 & -3 \end{bmatrix}^T$	0
$\begin{bmatrix} 4 & -1 \end{bmatrix}^T$	1

If we use a k-NN algorithm with $k=3$, what would be the predicted label for $x_{test} = [4 \ 0]^T$?

ANSWER

1 (Equal • Numeric)

CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider the following 2D data points along with its labels:

x	y
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 1 & -1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 2 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & -2 \end{bmatrix}^T$	0
$\begin{bmatrix} 3 & -3 \end{bmatrix}^T$	0
$\begin{bmatrix} 4 & -1 \end{bmatrix}^T$	1

Consider the following 2D data points along with its labels:

x	y
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 1 & -1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 2 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & -2 \end{bmatrix}^T$	0
$\begin{bmatrix} 3 & -3 \end{bmatrix}^T$	0
$\begin{bmatrix} 4 & -1 \end{bmatrix}^T$	1

Compute the Leave-one-out cross validation (LOOCV) error for the 3- nearest neighbour (3-NN) algorithm on this dataset. Enter the answer correct to two decimal places.

ANSWER

0.41 - 0.45 (Range - Numeric)

CONTEXT / PARAGRAPH

Based on the above data, answer the given subquestions.

Consider the following 2D data points along with its labels:

x	y
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 1 & -1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 2 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & -2 \end{bmatrix}^T$	0
$\begin{bmatrix} 3 & -3 \end{bmatrix}^T$	0
$\begin{bmatrix} 4 & -1 \end{bmatrix}^T$	1

Consider the following 2D data points along with its labels:

x	y
$\begin{bmatrix} 1 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 1 & -1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 1 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & 2 \end{bmatrix}^T$	1
$\begin{bmatrix} 2 & -2 \end{bmatrix}^T$	0
$\begin{bmatrix} 3 & -3 \end{bmatrix}^T$	0
$\begin{bmatrix} 4 & -1 \end{bmatrix}^T$	1

Is it possible to reduce the cross-validation error obtained in the previous question by changing the label of certain points?

OPTIONS

A. Yes ✓ **Correct** 23F3000704 SIDDHESH RAM K <23f3000704@ds.study.iitm.ac.in>

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B. No

C. Cannot conclude from the given information.

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